

$$\Delta v = 4 \text{ m/s}$$

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| 3 | D | <p>Initially, the speed of the car is increasing in the forward direction. <b>At X, the car reaches maximum speed (where acceleration is zero).</b></p> <p>Beyond X, as the acceleration is in the opposite direction to the velocity, the car slows down in speed at an increasing rate to Y then at a decreasing rate to Z; but the car keeps moving in the forward direction even at Z as the positive change in velocity from O to X (positive area under the <math>a-t</math> graph) is still greater than the negative change in velocity from X to Z (negative area under the <math>a-t</math> graph).</p> <p><b>Therefore, maximum displacement is at Z.</b></p> |
| 4 | B | The acceleration is downwards, thus resultant force is downwards.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |